

CS257 Linear and Convex Optimization

Homework 2

Due: September 28, 2020

September 21, 2020

1. Prove. If $C_1 \subseteq \mathbb{R}^m$ and $C_2 \subseteq \mathbb{R}^n$ are convex, so is $C_1 \times C_2 = \{(x, y) \in \mathbb{R}^{m+n} | x \in C_1, y \in C_2\}$.

Proof. If $z_1 = (x_1, x_2), x_1 \in C_1, x_2 \in C_2$, and $z_2 = (y_1, y_2), y_1 \in C_1, y_2 \in C_2$, then

$$\alpha z_1 + (1 - \alpha)z_2 = ((\alpha x_1 + (1 - \alpha)y_1), (\alpha x_2 + (1 - \alpha)y_2))$$

As C_1, C_2 are convex, $(\alpha x_1 + (1 - \alpha)y_1) \in C_1, (\alpha x_2 + (1 - \alpha)y_2) \in C_2$. □

2. Let $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ be an affine function from $\mathbb{R}^n$ to $\mathbb{R}^m$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Show that if $D \subset \mathbb{R}^m$ is convex, so is its inverse image $f^{-1}(D) \triangleq \{\mathbf{x} : f(\mathbf{x}) \in D\}$.

Proof. Let $\mathbf{x}_1, \mathbf{x}_2 \in f^{-1}(D)$ and $\theta \in [0, 1]$. By definition, $f(\mathbf{x}_i) \in D$. By convexity of D ,

$$f(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) = \mathbf{A}(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) + \mathbf{b} = \theta(\mathbf{A}\mathbf{x}_1 + \mathbf{b}) + \bar{\theta}(\mathbf{A}\mathbf{x}_2 + \mathbf{b}) = \theta f(\mathbf{x}_1) + \bar{\theta} f(\mathbf{x}_2) \in D$$

Thus

$$\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2 \in f^{-1}(D)$$

□

3. Show by induction on m that if $C \subset \mathbb{R}^n$ is convex and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in C$, then any convex combination $\sum_{i=1}^m \theta_i \mathbf{x}_i \in C$. (We have sketched the proof. Please fill in the details.)

Proof. If $m = 1$, trivial. If $m = 2$, the proposition is true by the definition of convex set. Assume the proposition is true for $m = k$. Consider the case $m = k + 1$. Let

$$\mathbf{x} = \sum_{i=1}^{k+1} \theta_i \mathbf{x}_i$$

- If $\theta_{k+1} = 1$, then $\mathbf{x} = \mathbf{x}_{k+1} \in C$.
- If $\theta_{k+1} \neq 1$, then

$$\mathbf{y} = \sum_{i=1}^k \frac{\theta_i}{1 - \theta_{k+1}} \mathbf{x}_i$$

is a convex combination of $\mathbf{x}_1, \dots, \mathbf{x}_k$ and hence belongs to C by the induction hypothesis. Now

$$\mathbf{x} = (1 - \theta_{k+1})\mathbf{y} + \theta_{k+1}\mathbf{x}_{k+1} \in C$$

This completes the induction step. Therefore, the proposition is true for any $m \in \mathbb{N}$. $\square$

4. Prove the convex hull $\text{conv } S$ of S is the set of all convex combinations of points in S by completing the following steps. You can assume the result in Problem 3.

(a). Let

$$C = \left\{ \sum_{i=1}^m \theta_i \mathbf{x}_i : m \in \mathbb{N}; \mathbf{x}_i \in S, \theta_i \geq 0, i = 1, \dots, m; \sum_{i=1}^m \theta_i = 1 \right\}$$

Show that C is convex.

(b). Show that $C \subset \text{conv } S$ and conclude $C = \text{conv } S$.

Proof. (a). Let $\mathbf{y}_1, \mathbf{y}_2 \in C$ and $\alpha \in [0, 1]$. Then

$$\mathbf{y}_1 = \sum_{i=1}^{m_1} \theta_{1,i} \mathbf{x}_{1,i}, \quad \mathbf{y}_2 = \sum_{i=1}^{m_2} \theta_{2,i} \mathbf{x}_{2,i}$$

where $\mathbf{x}_{i,j} \in S$, $\theta_{i,j} \geq 0$, $i = 1, 2, j = 1, 2, \dots, m_i$ and $\sum_{j=1}^{m_i} \theta_{i,j} = 1$, $i = 1, 2$. Consider

$$\alpha \mathbf{y}_1 + \bar{\alpha} \mathbf{y}_2 = \sum_{i=1}^{m_1} \alpha \theta_{1,i} \mathbf{x}_{1,i} + \sum_{i=1}^{m_2} \bar{\alpha} \theta_{2,i} \mathbf{x}_{2,i}.$$

Since $\alpha \theta_{1,i} \geq 0$, $\bar{\alpha} \theta_{2,i} \geq 0$, and

$$\sum_{i=1}^{m_1} \alpha \theta_{1,i} + \sum_{i=1}^{m_2} \bar{\alpha} \theta_{2,i} = \alpha + \bar{\alpha} = 1,$$

$\alpha \mathbf{y}_1 + \bar{\alpha} \mathbf{y}_2$ is a convex combination of $\mathbf{x}_{1,1}, \dots, \mathbf{x}_{1,m_1}, \mathbf{x}_{2,1}, \dots, \mathbf{x}_{2,m_2}$ and hence belongs to C . Therefore C is convex.

(b). Since $\text{conv } S$ is convex, it contains all convex combinations of points in S , so $C \subset \text{conv } S$. Since C is convex, $C = \text{conv } S$ by the minimality of $\text{conv } S$. $\square$

5. Let $\mathbf{x}_0, \dots, \mathbf{x}_K$ be distinct points in $\mathbb{R}^n$. Let V be the set of points that are closer in Euclidean distance to $\mathbf{x}_0$ than $\mathbf{x}_1, \dots, \mathbf{x}_K$, called the **Voronoi region** around $\mathbf{x}_0$ with respect to $\mathbf{x}_1, \dots, \mathbf{x}_K$, i.e.

$$V = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2, i = 1, 2, \dots, K\}.$$

Show that V is a polyhedron by identifying $\mathbf{A}$ and $\mathbf{b}$ such that $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$. (You don't have to draw it in the submission, but try to visualize V for $\mathbb{R}^2$.)

Proof. Note that

$$\|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2 \iff \|\mathbf{x} - \mathbf{x}_0\|_2^2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2^2$$

$$\begin{aligned}
&\Longleftrightarrow (\mathbf{x} - \mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) \leq (\mathbf{x} - \mathbf{x}_i)^T (\mathbf{x} - \mathbf{x}_i) \\
&\Longleftrightarrow -2\mathbf{x}_0^T \mathbf{x} + \mathbf{x}_0^T \mathbf{x}_0 \leq -2\mathbf{x}_i^T \mathbf{x} + \mathbf{x}_i^T \mathbf{x}_i \\
&\Longleftrightarrow 2(\mathbf{x}_i - \mathbf{x}_0)^T \mathbf{x} \leq \mathbf{x}_i^T \mathbf{x}_i - \mathbf{x}_0^T \mathbf{x}_0
\end{aligned}$$

Let

$$\mathbf{A} = \begin{pmatrix} 2(\mathbf{x}_1 - \mathbf{x}_0)^T \\ \vdots \\ 2(\mathbf{x}_K - \mathbf{x}_0)^T \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \mathbf{x}_1^T \mathbf{x}_1 - \mathbf{x}_0^T \mathbf{x}_0 \\ \vdots \\ \mathbf{x}_K^T \mathbf{x}_K - \mathbf{x}_0^T \mathbf{x}_0 \end{pmatrix}$$

Then $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$, so V is a polyhedron. □